Inverse Function Learning for Image Deblurring: Classical vs. Modern Machine Learning

Surya Manoj Pentakota
2026701036, IIIT Hyderabad

Kevin Paul Reddy Janagari
2026900006, IIIT Hyderabad

***Abstract*—Image deblurring is one of the most common ill-posed inverse problems due to its instability and non-unique solution to it, the blur operator A diminishes high-frequency content, so no single stable inverse exists and naively inverting it, the noise will be amplified exponentially resulting in a completely distorted image. We frame deblurring as *learning inverse functions*, find g_θ minimizing $\mathbb{E}\|g_\theta(y) - x\|^2$ over (x, y) pairs from the known degradation model, and study the central design question: should g_θ be built around the physics of A, or purely from data? We propose a controlled comparison of two methods: (M1) a modern, data-only CNN that receives no information about the blur kernel or noise, and (M2) a classical-ML iterative solver with a learned shallow prior (EPLL/dictionary) that is given the true kernel and noise level. To isolate the effect of physics awareness in the (M2) model, we propose to use a kernel-aware unrolled network (USRNet-style). We will evaluate on synthetic data (BSDS) and the GoPro benchmark, measuring PSNR, SSIM, LPIPS, data-consistency, kernel-mismatch robustness, and adversarial stability, with paired statistical tests.**

I. Introduction

A sharp image x is observed through blur and noise:

$$y = Ax + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where A is convolution with a point-spread-function (PSF) kernel k and ϵ is sensor noise. Because A is ill-conditioned its eigenvalues decay to near zero at high frequencies therefore blurring destroys information and naive inversion amplifies noise into unusable output [1], [2]. The classical approaches regularizes the inversion with a hand-crafted prior such as total variation (TV) [1], solved iteratively with ISTA/FISTA [3].

Goal. We reformulate deblurring as *inverse function learning*: learn $g_\theta : \mathcal{Y} \to \mathcal{X}$ that minimizes the expected reconstruction error $\mathbb{E}\|g_\theta(y) - x\|_2^2$. Every existing method is a different answer to how g_θ should be constructed, and the methods differ along one axis: *how much the model knows about A*. Classical solvers consume the kernel and noise model; modern deep networks learn the inverse from data alone [4], [5]; deep unfolding (LISTA [6], ISTA-Net [7], USRNet [8]) keeps A inside a learned network.

Planned approach. We design a controlled study comparing (M1) a modern, data-only CNN trained on (blurry, sharp) pairs with no access to k or σ; (M2) a classical-ML pipeline, an iterative ADMM/HQS solver with a learned shallow prior (Expected Patch Log Likelihood [9], optionally K-SVD dictionaries [10]) given the true k and σ; and a control: a kernel-aware unrolled network [8] that receives k and σ as inputs. This isolates what each component physics-awareness, prior depth, learning contributes.

Contributions.

- A controlled cross-family comparison of inverse-function learners for deblurring under one degradation protocol (shared PSF library, noise grid, boundary handling, test set).
- Quantitative kernel-mismatch robustness curves (train on PSF set P_1, sweep test PSFs): direct CNNs are known to collapse under PSF shift [11]; we measure the gap against physics-aware methods.
- An in-domain stability test for deblurring, transferring the instability analysis of [12] to the convolution regime.
- An unrolling-genealogy ablation (LISTA → ISTA-Net → USRNet) isolating the contribution of each learned component.

II. How Current Approaches Work

Classical iterative solvers. ISTA/FISTA [3] alternate a gradient step on the data term $\frac{1}{2}\|Ax - y\|_2^2$ with a proximal step enforcing a hand-crafted prior (TV [1]). They require k and σ, take hundreds to thousands of iterations, and TV's piecewise-constant prior produces staircase artifacts on natural texture [3].

Deep unfolding. LISTA [6] unrolls ISTA to a fixed depth and learns the iteration matrices from data, reducing iterations roughly 20× at equal accuracy. ISTA-Net [7] replaces the hand-crafted proximal step with a learned CNN transform. USRNet [8] unfolds half-quadratic splitting with a closed-form FFT data module and a ResUNet denoiser, taking the blur kernel and noise level as inputs so one model serves arbitrary degradations.

Classical-ML priors. Fields of Experts [13] learn MRF filters from data; EPLL [9] learns a 200-component Gaussian mixture patch prior and was the strongest generic prior for deblurring at the time; K-SVD [10] learns overcomplete dictionaries for sparse coding. All remain inside iterative solvers and need k.

Direct deep regression. DeepDeblur [4] and NAFNet [5] (33.69 dB on GoPro) learn kernel-free mappings from

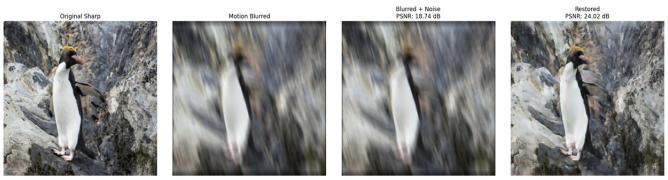


Fig. 1. Results with Deep Unfolding with physics provided. This serves as the baseline for our approaches

TABLE I
Difference Between the Two Proposed Methods

	M1: Modern (data-only)	**M2: Classical (physics-aware)**
Knows blur kernel	No (implicit in data)	Yes (explicit input)
Knows noise model	No	Yes
Inversion	Single learned pass	Iterative optimization
Prior	Implicit deep CNN	Explicit shallow (GMM/sparsity)
Inference time	∼ms per image	seconds-minutes
Expected failure	Kernel mismatch, hallucination, instability [12]	Staircase/ringing artifacts; needs correct kernel

real blurry/sharp pairs; the degradation is only implicit in the data. Deep Image Prior [14] uses an untrained network as a zero-shot prior.

Gap. No existing study compares these families under one controlled protocol; results are fragmented across papers with different kernels, noise, and evaluation. Moreover, direct networks are known to fail under kernel mismatch [11], and learned inverses are provably unstable to tiny perturbations [12] yet neither effect has been quantified systematically for deblurring.

III. Proposed Approach

Method 1 (Modern ML, data-only). g_θ is a CNN trained purely on (x, y) pairs sampled from (1); the network never receives k or σ, so it must infer both implicitly (blind by construction). Architecture: a U-Net regression baseline, optionally extended with NAFNet-style blocks [5].

Method 2 (Classical ML, physics-aware). g_θ is an iterative ADMM/HQS solver with a *learned shallow* prior EPLL (GMM) [9] or K-SVD dictionaries [10] given the true kernel and noise level. The solver performs the inversion; the prior performs the regularization.

Control (Unrolled). A USRNet-style network [8] receives k and σ as inputs. It is the "bridge" family: physics inside, prior learned. Including it lets us attribute any M1–M2 gap to either the prior or the physics-awareness, rather than confound the two.

Table I summarizes the design differences that drive our hypotheses H1–H3: (H1) the M1-M2 gap is small at low noise with the correct kernel and widens with noise, texture complexity, and kernel mismatch; (H2) on piecewise-constant content M2 is competitive, while on natural texture M1 wins; (H3) M1 is unstable under tiny perturbations while physics-aware methods resist them [12].

IV. Evaluation Plan

Data. Tier 1 (core): BSDS300/500 [15] clean images, degraded synthetically via (1) with circular (FFT) boundary handling. PSF library: Gaussian kernels ($\sigma_k \in \{0.7, 1.2, 1.6, 2.0\}$), motion kernels (Levin benchmark [2]), defocus disks. Noise $\sigma \in \{0, 1, 5, 15, 25\}/255$. Tier 2 (stretch): the GoPro dataset [4] (3214 real blurry/sharp pairs, kernels not provided) for blind deblurring.

Metrics. Reconstruction quality: PSNR, SSIM, LPIPS. Fidelity to the forward model: data-consistency residual $\|A\hat{x} - y\|_2$. For Tier 2: Levin kernel-error ratio [2]. Robustness: PSNR drop under kernel mismatch (H1) and under worst-case perturbations (H3, projected gradient ascent per [12]). All paired comparisons tested with the Wilcoxon signed-rank test across test images; seeds fixed for reproducibility.

Success criteria. H1-H3 confirmed or refuted with statistical significance; the cross-family results table, kernel-mismatch curves, and stability figure; a reproducible protocol released with the report.

V. Team Work Split

Both members jointly design the protocol, cross-verify each other's numbers, and integrate results weekly. The split is organized along the classical/modern axis, so each member owns a complete pipeline end-to-end.

TABLE II
Work Split Between the Two Team Members

Student One	**Student Two**
Degradation pipeline, PSF library, dataset splits	Unrolled networks (LISTA, ISTA-Net, USRNet)
Classical baselines: TV-ADMM, FISTA, Wiener, Richardson–Lucy	Direct CNN (M1) and Deep Image Prior
Classical-ML priors: EPLL, K-SVD (M2)	Kernel-mismatch sweep and instability test (H1/H3)
Metrics harness and statistical tests	Tier 2 GoPro experiments
Report figures/tables, final write-up (joint)	Report write-up, 5-min video

VI. Compute

We have access to a shared NVIDIA GPU with [RTX 3090, VRAM 24GB] and internet access for dataset download (BSDS500, GoPro Large [4]). Estimated budget: Tier 1 synthetic experiments (128×128 patches, five families) ≈ [10–20] GPU-hours; Tier 2 fine-tuning on GoPro ≈ [5–10] GPU-hours; Deep Image Prior [14] is run per-image on a small test subset only. All classical baselines run on CPU within minutes per image. Total compute fits within a single semester with margin; if GPU time is contested, we first reduce epochs/patch counts, and the classical methods remain fully executable CPU-only.

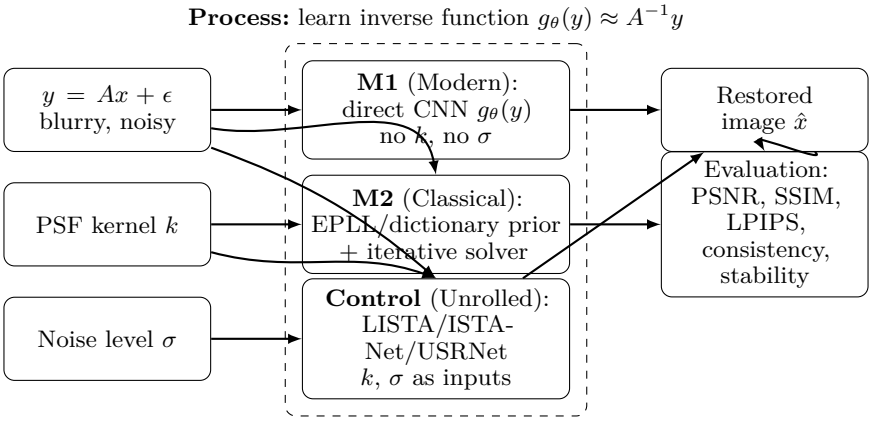


Fig. 2. Overview: Inputs → Process/Approach → Output. All methods take the blurry image y; M2 and the control additionally consume the kernel k and noise level σ, while M1 uses none.

VII. Timeline and Milestones

TABLE III
Project Timeline (12-week semester)

Phase	Weeks	Deliverables
1	1–2	Degradation pipeline, PSF library, dataset splits, repo skeleton
2	3–4	Classical baselines (TV-ADMM, FISTA, Wiener, RL); metrics harness
3	5–6	Classical-ML priors (EPLL, K-SVD); preliminary H1/H2 on classical side
4	7–8	Unrolled networks (LISTA, ISTA-Net, USRNet) + direct CNN (M1)
5	9–10	Kernel-mismatch curves (H1) and instability test (H3)
6	11	Tier 2 blind deblurring on GoPro (stretch goal)
7	12	Final report, 5-minute video, presentation

VIII. Conclusion

The project delivers a controlled, statistically tested map of when classical physics-aware ML is sufficient and when modern learned inverses are needed and what the latter cost in robustness.

References

[1] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D: Nonlinear Phenomena*, vol. 60, no. 1–4, pp. 259–268, 1992.

[2] A. Levin, Y. Weiss, F. Durand, and W. T. Freeman, "Understanding and evaluating blind deconvolution algorithms," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 1964–1971.

[3] A. Beck and M. Teboulle, "Fast gradient-based algorithms for constrained total variation image denoising and deblurring problems," *IEEE Transactions on Image Processing*, vol. 18, no. 11, pp. 2419–2434, 2009.

[4] S. Nah, T. H. Kim, and K. M. Lee, "Deep multi-scale convolutional neural network for dynamic scene deblurring," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[5] L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022.

[6] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 2010, pp. 399–406.

[7] J. Zhang and B. Ghanem, "Ista-net: Interpretable optimization-inspired deep network for image compressive sensing," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 1828–1837.

[8] K. Zhang, L. Van Gool, and R. Timofte, "Deep unfolding network for image super-resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3217–3226.

[9] D. Zoran and Y. Weiss, "From learning models of natural image patches to whole image restoration," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 479–486.

[10] M. Aharon, M. Elad, and A. Bruckstein, "K-svd: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Transactions on Signal Processing*, vol. 54, no. 11, pp. 4311–4322, 2006.

[11] K. Zhang, W. Zuo, S. Gu, and L. Zhang, "Learning deep CNN denoiser prior for image restoration," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3929–3938.

[12] V. Antun, F. Renna, C. Poon, B. Adcock, and A. C. Hansen, "On instabilities of deep learning in image reconstruction and the potential costs of AI," *Proceedings of the National Academy of Sciences*, vol. 117, no. 48, pp. 30 088–30 095, 2020.

[13] S. Roth and M. J. Black, "Fields of experts: A framework for learning image priors," in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, vol. 2, 2005, pp. 860–867.

[14] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Deep image prior," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[15] P. Arbeláez, M. Maire, C. Fowlkes, and J. Malik, "Contour detection and hierarchical image segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 5, pp. 898–916, 2011.

IX. Video Proposal

Please check the video proposal at the link below:
https://drive.google.com/file/d/1KEAgSBaRigIuQ0v0uQXqFWGOQSC7pQMY/view?usp=sharing